ASIAN INSTITUTE OF MANAGEMENT

Data Mining and Wrangling

Web Scraping with Python

Session 11

BSDSBA 2028

04 March 2026

ASIAN INSTITUTE OF MANAGEMENT

Class Administrative Matters

Course Deliverables

Assessed – R05, E2S2 (Optional), ~~ICA04, ICA05, E3~~

Due this Week – R06 (Wed EOD)

Upcoming – Midterm Exam, E3S2, ICA06, E4, A1, Mini Project 1

Supplementary Materials

Git branching tutorial

Sample midterms

Class Administrative Matters

Midterm Exam

Duration: 3 hours

Composition: Multiple Choice 1pt each (30 items), Problem Solving 5 pts each (5 items), Offline Coding 5 pts each (2 items). **Total: 65 pts**

Venue: Fuhler Hall (TPIC and Urbano)

Time: 11:00 AM to 2:00 PM

Mechanics: Only pens/pencil allowed, RegEx cheat sheet will be provided, scratch paper will be provided, no smartphones and other gadgets allowed, limited bathroom breaks.

Class Administrative Matters

Other

Course Achievements and Badges

Session 10 and 11 – Web Scraping with Python

Gameplan

Session 10

Fundamentals of Web Scraping

Web Scraping using BeautifulSoup

Session 11

API Fundamentals

APIs for Data Gathering

AIM

Fundamentals of Web Scraping

Fundamentals of Web Scraping

What is web scraping?

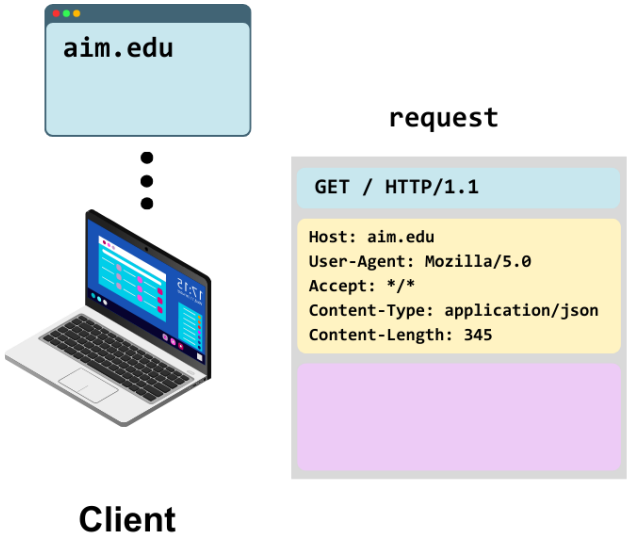
Web scraping is the process of extracting data from websites programmatically. It generally involves automating repetitive tasks such as clicking buttons or traversing links on a webpage.

In essence, web scraping is substituting a web browser for an application of your design. Thus, allowing you to collect, store, and analyze data from websites efficiently.

Fundamentals of Web Scraping

How do websites work?

The web operates on a **client-server model**, where your computer (the client) communicate with a website's server using **HTTP (Hypertext Transfer Protocol)**.



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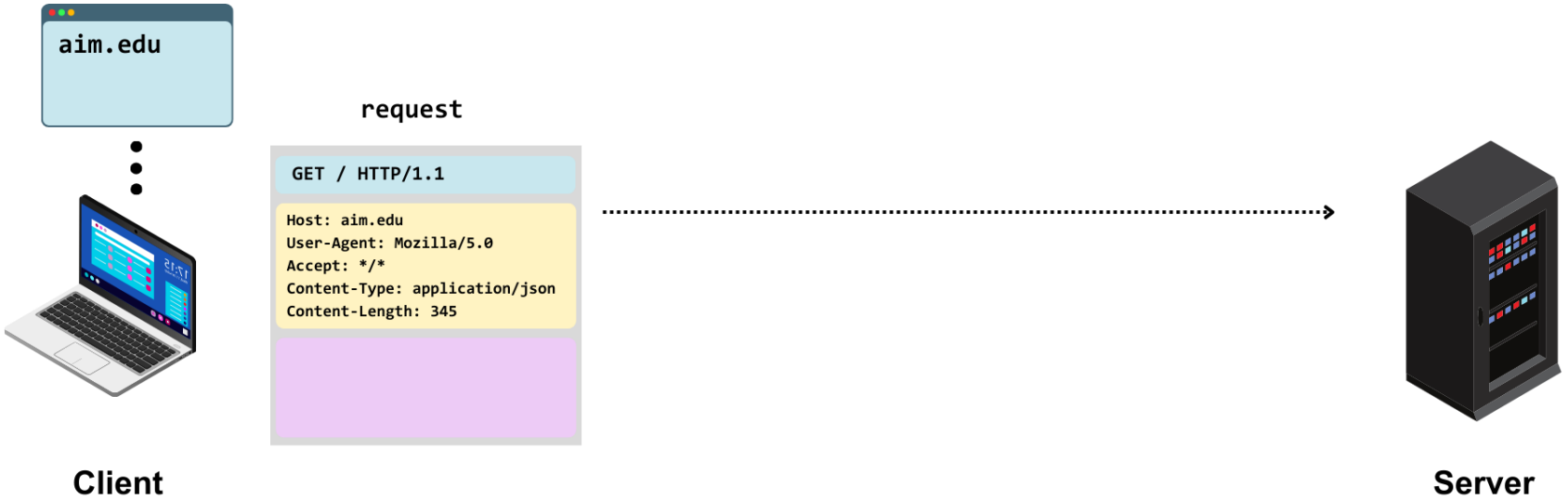

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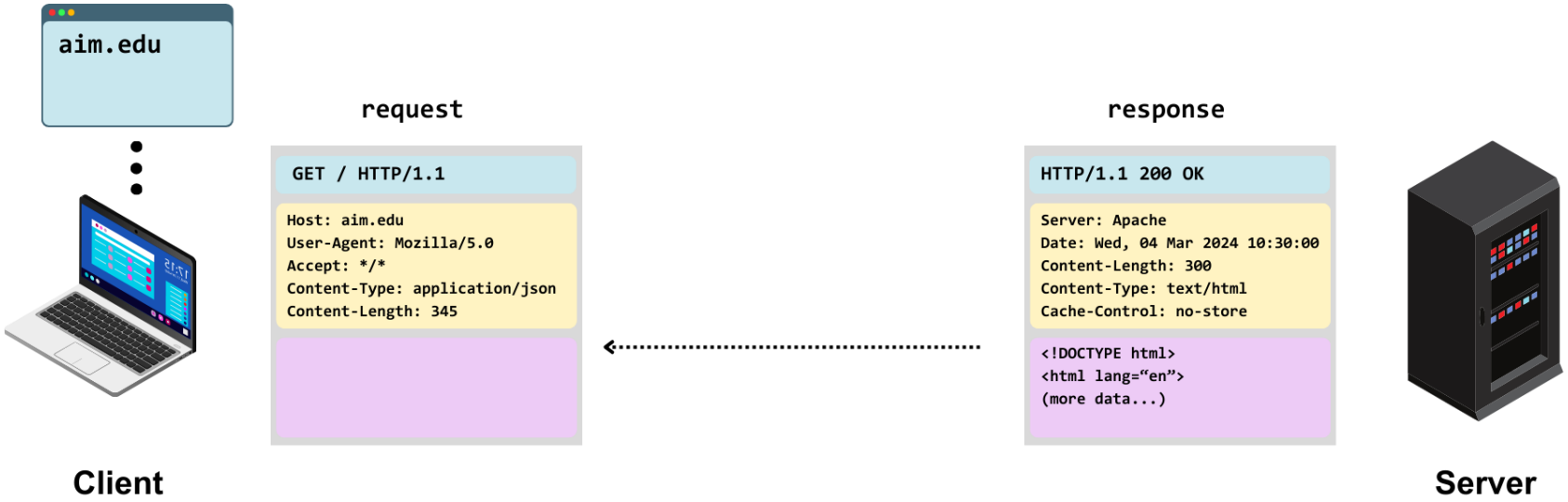

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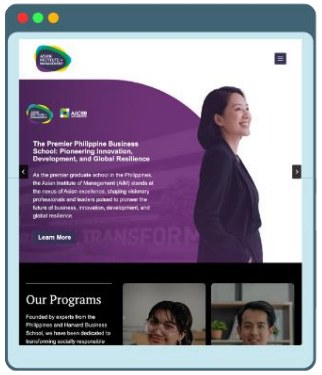
Client



response

```
HTTP/1.1 200 OK

Server: Apache
Date: Wed, 04 Mar 2024 10:30:00
Content-Length: 300
Content-Type: text/html
Cache-Control: no-store

<!DOCTYPE html>
<html lang="en">
(more data...)
```

Server

Fundamentals of Web Scraping

What are the different types of requests?

- **GET**: Most common method for requesting information. (e.g., loading a webpage)
- **POST**: Used to submit information to the server in a secure way. (e.g., submitting a login form)
- Others: **PUT**, **DELETE**, etc., are less commonly used in scraping but used in APIs.

Fundamentals of Web Scraping

What are the different response status codes?

- **200**: OK, request succeed
- **404**: Not found, requested resource cannot be found
- **403**: Access denied (e.g., blocked by the server)
- **429**: Too many requests (rate-limited)
- **500**: Internal server error

Fundamentals of Web Scraping

What do websites usually contain?

- **HTML (HyperText Markup Language) –** The standard markup language used for building a web page.
- **CSS (Cascading Style Sheets)** – Define the appearance of HTML elements on a web page.
- **JavaScript** (also known as **JS** or **JScript**) – “client-side” code used for adding dynamic features and providing user-based interaction inside web pages.

Fundamentals of Web Scraping

HTML Basics

```
<html>
  <head>
    <title> A Simple Webpage </title>
  </head>
  <body>
    <h1> Welcome, Friends! </h1>
    <p> This is a paragraph </p>
    <a href="https://example.com">An Example Link</a>
  </body>
</html>
```

- **Tags** – define the elements of the HTML document. (e.g., <a>, <h1>, <p>).
- **Attributes** – provides additional information to the element (e.g., href, src, id, class).
- **Hierarchy** – elements can be nested in a tree-like structure with a parent-child hierarchy.

Fundamentals of Web Scraping

CSS Selectors Basics

```
h1#main-title {
  font-size: 20px;
}
div.content {
  color: green;
}
.content[data="1.5"] {
  color: red;
}
```

- `tag` – Selects all `<tag>` elements
- `.class` – Selects elements with `class="class"`
- `#id` – Selects elements with `id="id"`
- `[attribute="value"]` – Selects elements with specified attribute values

Fundamentals of Web Scraping

JavaScript Basics

```
<script>


// List of words
var myWords = [
   "Hello", "Everybody", "How", "Are", "You", "Today",
   "It", "Is", "A", "Lovely", "Day", "I", "Love",
   "Coding", "In", "My", "Van", "Mate”
]
...
</script>
```

- Embedded in the HTML using the `<script>` tag, can be inserted as a content or in a separate file.
- You won’t likely need to read or write JavaScript, but it’s useful to know what it looks like.
- Scripts can also contain useful data.

Fundamentals of Web Scraping

Web Scraping Tools

- **Developer Tools –** Shows how websites have been put together.
- `requests` – Sends HTTP request to fetch webpage content.
- `BeautifulSoup` (`bs4`) – Parses HTML documents to extract specific elements.
- `selenium` – Allows interactive control with dynamic websites.

Fundamentals of Web Scraping

Key Principles of Ethical Web Scraping

- **Respect `robots.txt` –** This text file specifies which pages bots can access (and hence be scraped).
- **Don't Overload Servers –** Sending too many requests too quickly can crash a site.
- **Use Public Data** – Only scrape publicly available data, not private or login-protected content.
- **Follow Terms of Service** – Stay compliant to websites' ToS.
- **Protect Data Privacy** – Don't collect personal data (e.g., names, emails) without consent.

Fundamentals of Web Scraping

Best Practices of the Ethical Scraper

- Identify yourself in the requests with a **user agent** (e.g., “`MyScraper/1.0 (contact: your.email@example.com)`”).
- Use APIs when available, as they’re designed for data access.
- Always add delays in between requests (typically 1 to 2 seconds in between)
- Test on small scales first to avoid unintended server stress.
- Keep records of what you scrape and why, especially for commercial projects.

Fundamentals of Web Scraping

Exercise 1

- Choose a website you're interested in scraping.
- Find and review the site's `robots.txt` to see which parts of the site are allowed or disallowed for scraping.
- Record your findings onto the google sheet – The Ethical Web Scraper

Web Scraping using BeautifulSoup

Web Scraping with BeautifulSoup

BeautifulSoup Features

Finding tags

```
soup.find("tag")
soup.find_all("tag")
```

Accessing tags and attributes

```
soup.tag
soup.tag["attribute"]
```

Navigating the HTML Tree

- **children** – `soup.tag.children`
- **parent** – `soup.tag.parents`
- **siblings** – `soup.tag.next_siblings`

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What is an API?

An **Application Program Interface (API)** defines a standardized syntax that allows a piece of software to communicate with another piece of software, even though they might be written in different languages or otherwise structured differently.

User / Application

API

Fundamentals of Web Scraping

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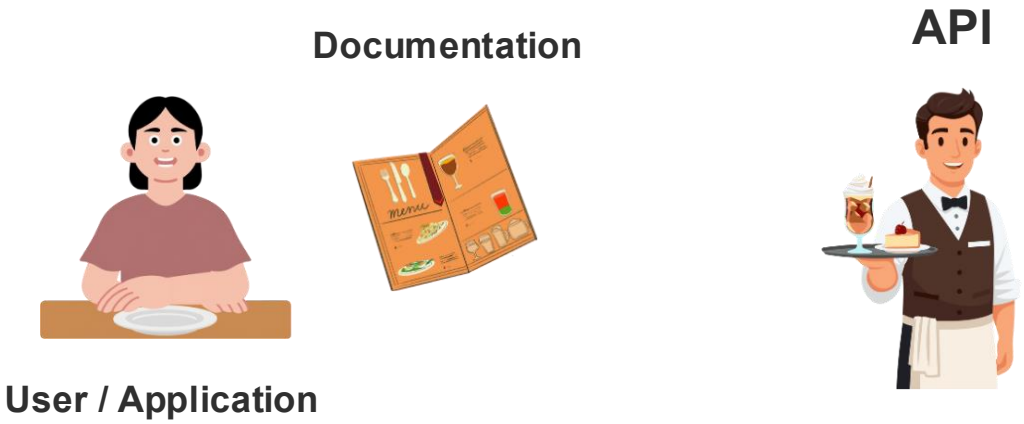
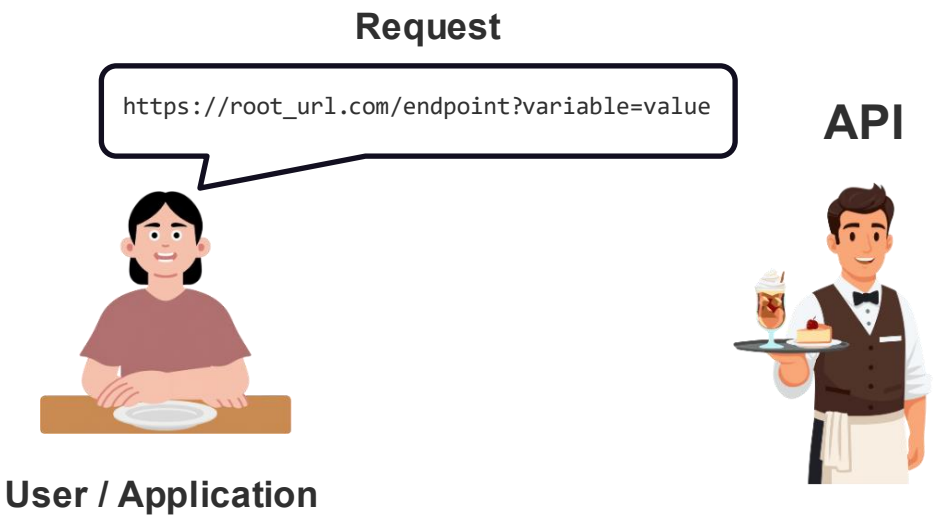


Fundamentals of Web Scraping

How does an API work?



Fundamentals of Web Scraping

How does an API work?

User / Application

API

Server

Fundamentals of Web Scraping

How does an API work?

API

User / Application

Data / Response

Server

Fundamentals of Web Scraping

How does an API work?

```
{
  "query": "value",
  "status": "success",
  "key_1": "value_1",
  "key_2": "value_2"
}
```

API

User / Application

Data / Response

Server

Fundamentals of Web Scraping

How does an API work?

1. A **client** identifies a need for **data** or **functionality**.
2. The client consults the **API documentation**.
3. The client constructs and sends a request to the **API endpoint**.
4. The API receives and validates the **request**.
5. The API forwards the valid request to the **backend server** or service.
6. The backend **processes** the requests.
7. The backend returns the **result** to the API.
8. The client receives and uses the **response**.

Fundamentals of Web Scraping

Why use an API?

- API provides structured data eliminating the need for HTML parsing.
- APIs are often faster and more reliable than scraping.
- APIs are generally created and supported by website owners, reducing ethical/legal concerns

Fundamentals of Web Scraping

Best Practices when using an API

- Read the API documentation thoroughly.
- Use environment variables for API keys.
- Be aware of rate limits.
- Test with small data sets first then scale up.

Reminders

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